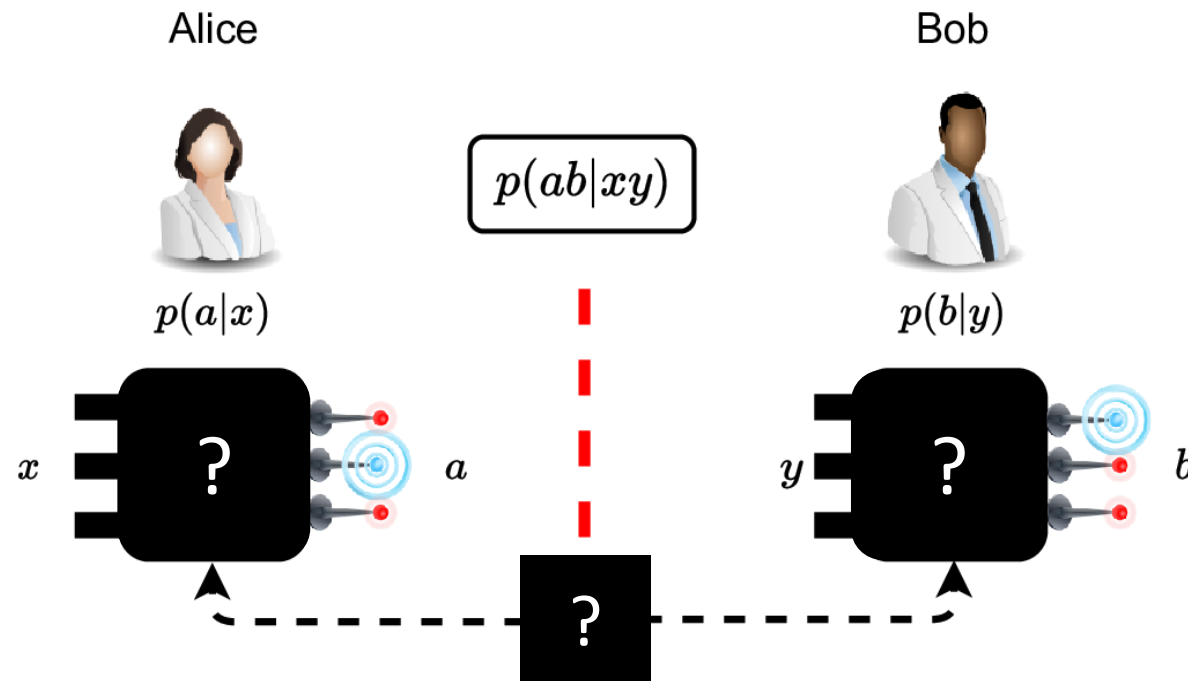
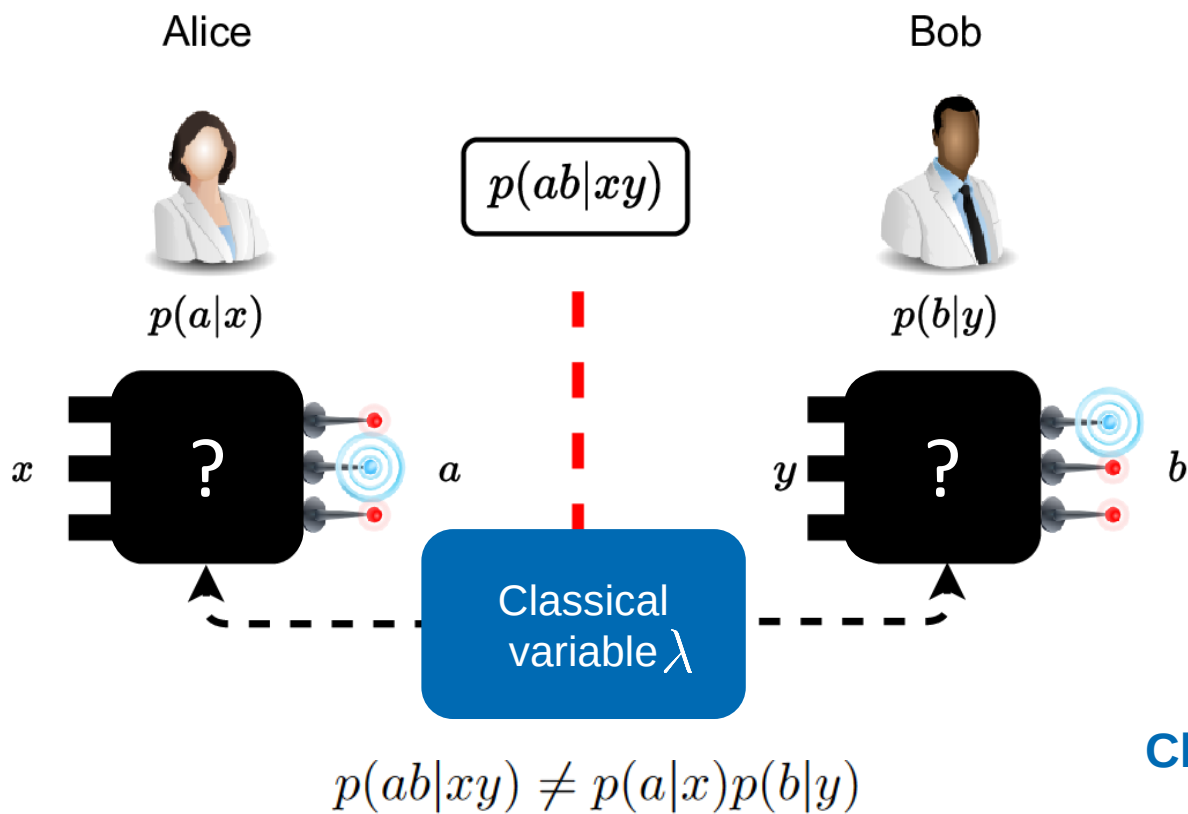


Applications of quantum nonlocality, review on related open problems in quantum information theory

Lucas Tendick



Basics of the Bell scenario



- No-communication scenario
- Prior agreement to a strategy is possible
- Questions/Inputs are chosen randomly (often also independently)
- No reference to inner functioning of the box
- Run of an experiment over many rounds

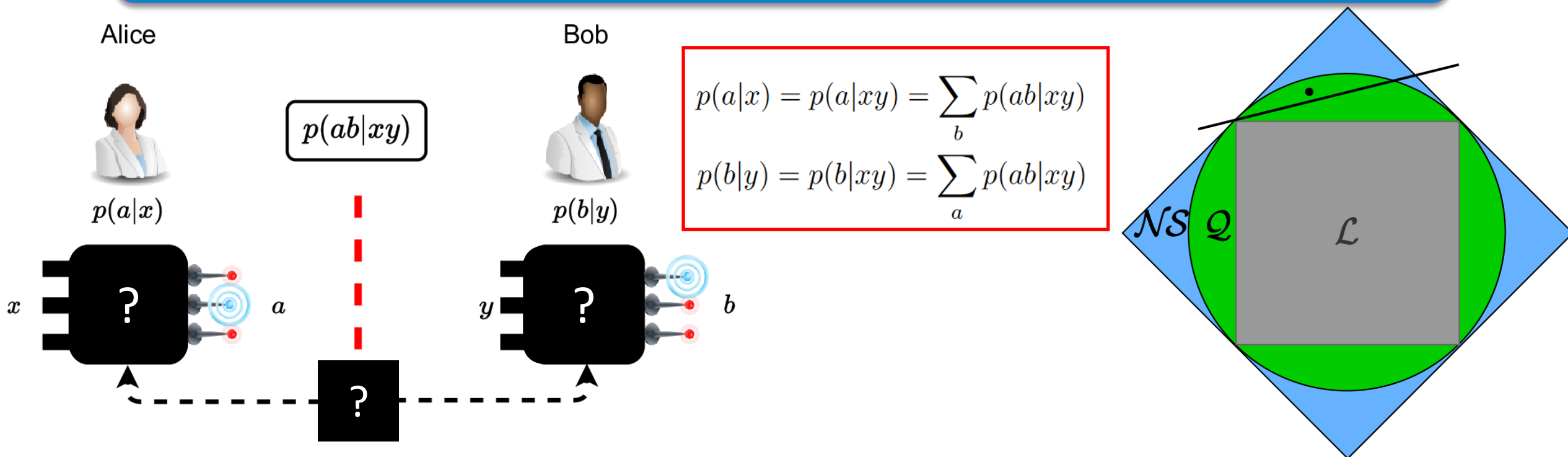
Classical Ansatz: $p(ab|xy, \lambda) = p(a|x, \lambda)p(b|y, \lambda)$

How to explain the correlations?

Local:

$$p(ab|xy) \stackrel{\mathcal{L}}{=} \sum_{\lambda} p_{\lambda} p(a|x, \lambda) p(b|y, \lambda)$$

Basic Quantum Advantages



Bell inequality:
$$\sum_{a,b,x,y} C_{ab|xy} p(ab|xy) \stackrel{\mathcal{L}}{\leq} L \stackrel{\mathcal{Q}}{\leq} Q \stackrel{\mathcal{NS}}{\leq} \text{NS}$$

CHSH inequality:
$$\langle A_1 B_1 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_1 \rangle - \langle A_2 B_2 \rangle \stackrel{\mathcal{L}}{\leq} 2 \stackrel{\mathcal{Q}}{\leq} 2\sqrt{2} \stackrel{\mathcal{NS}}{\leq} 4$$

with $\langle A_x B_y \rangle = p(a = b|xy) - p(a \neq b|xy)$

CHSH Game:

$$a \oplus b = x \cdot y$$

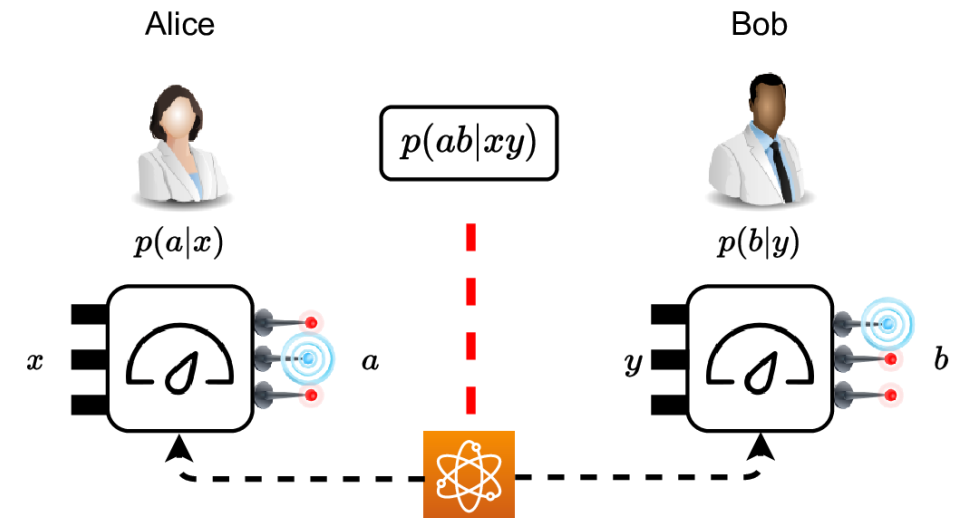
$$L = \frac{3}{4} \quad Q \approx 0.85 \quad \text{NS} = 1$$

Quantum Theory

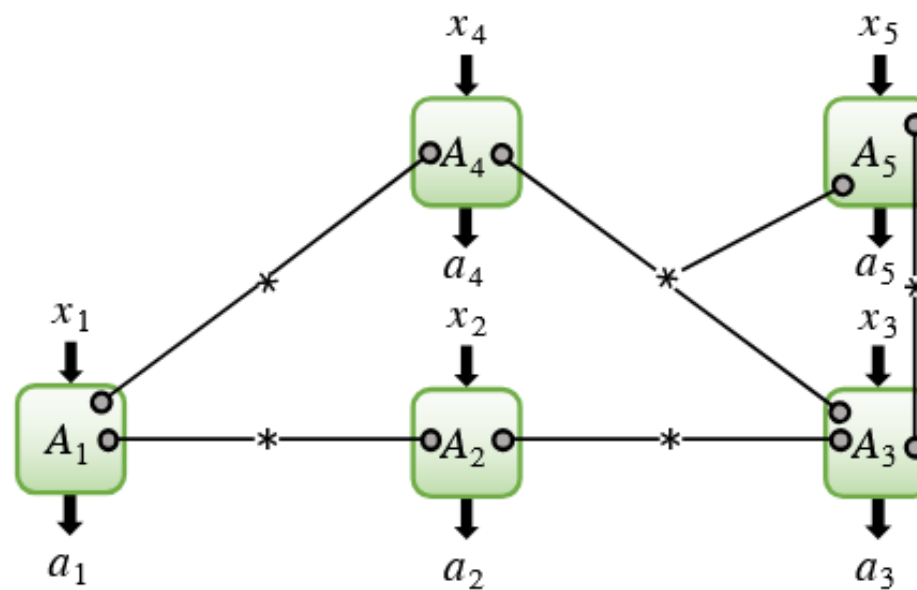
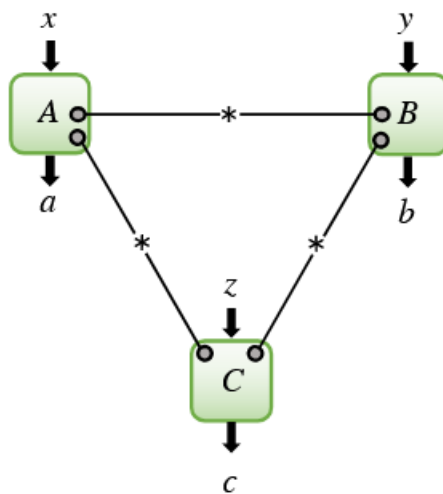
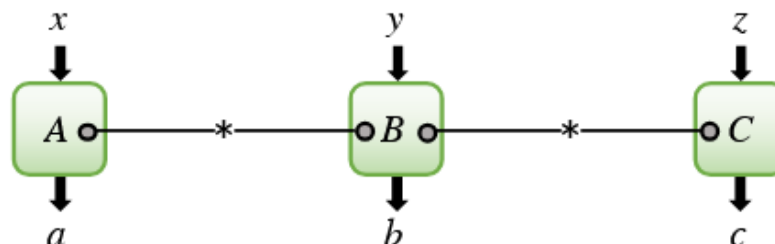
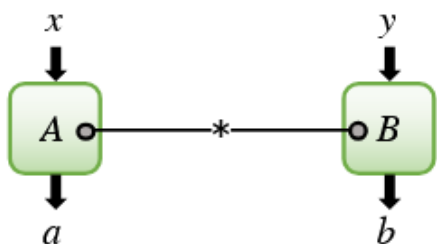
Quantum theory 101:

1. **States in complex Hilbert space** $|\Psi\rangle \in \mathcal{H} \longrightarrow \rho = \sum_i p_i |\Psi_i\rangle\langle\Psi_i| \iff \rho \geq 0, \text{Tr}[\rho] = 1$
2. **Evolution of systems** $\rho \longmapsto \rho' = U\rho U^\dagger$
3. **Measurements** $\{M_a\}_a$ s.t. $0 \leq M_a \leq \mathbb{1}, \sum_a M_a = \mathbb{1} \longrightarrow p(a) = \text{Tr}[M_a \rho]$
4. **Composite systems** $\mathcal{H}_{ST} := \mathcal{H}_S \otimes \mathcal{H}_T \longrightarrow \rho_{ST} := \rho_S \otimes \rho_T$

Quantum correlations: $p(ab|xy) = \text{Tr}[(A_{a|x} \otimes B_{b|y})\rho]$

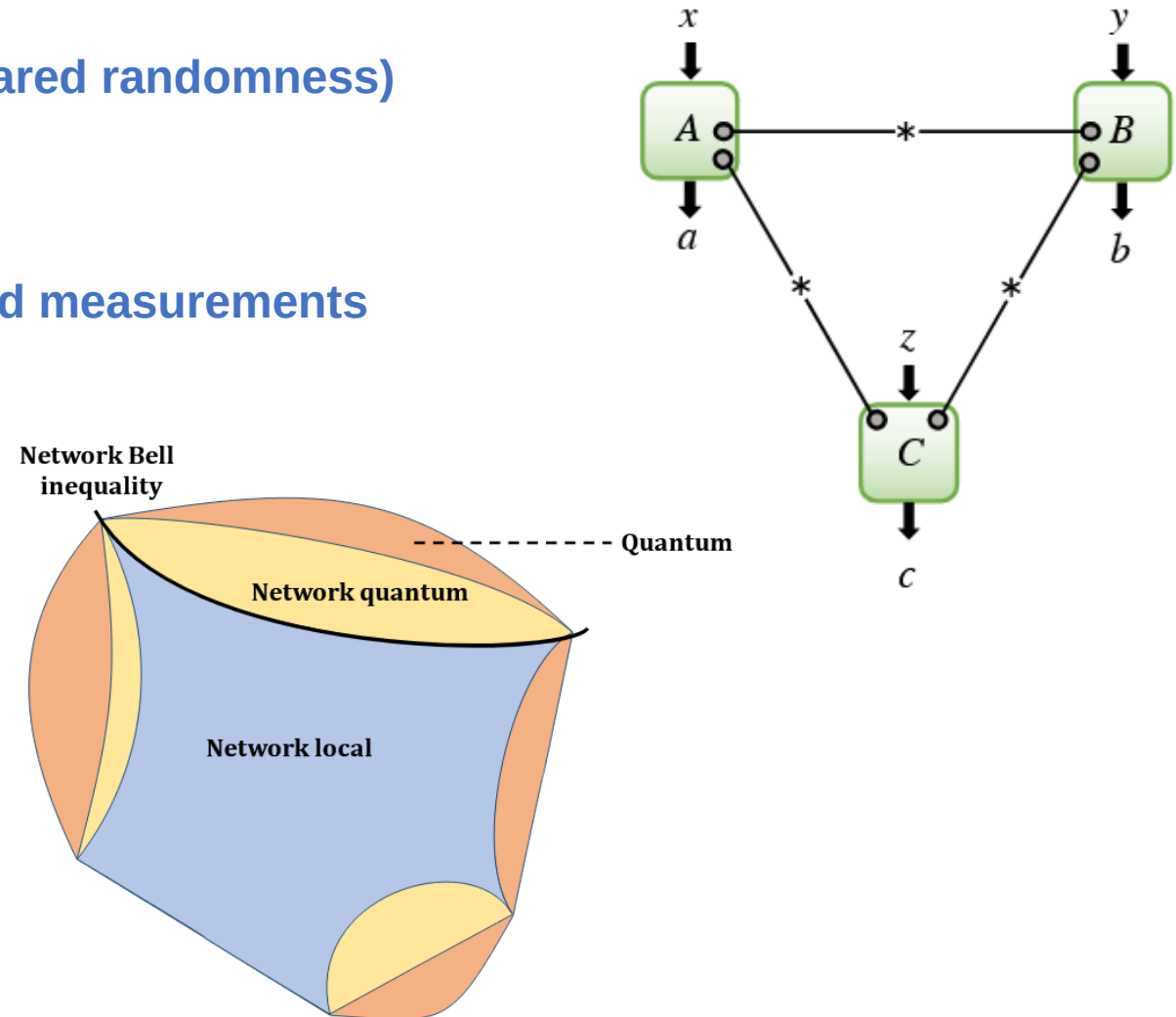


Network Nonlocality



Basics of Network Nonlocality

- Independent sources (could be assisted by shared randomness)
- No settings required for quantum advantages
- Usually thought of as a set-up with sources and measurements
- Typically no transformations in between
- Sets of correlations are not convex anymore
- Less generic tools available so far

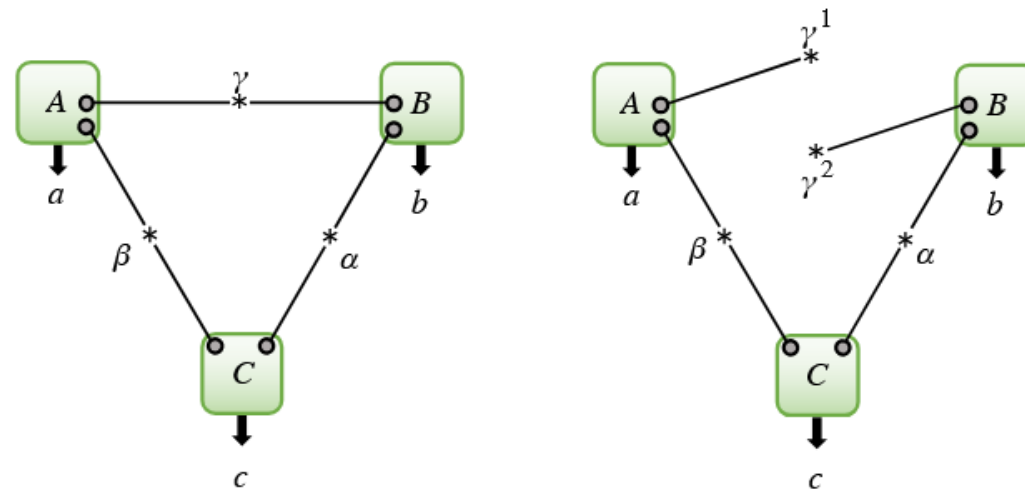


No-go Results in Networks

Can we achieve a shared random bit in the triangle network?

$$P_{\text{GHZ}} := \frac{1}{2}([000] + [111])$$

The Proof does not rely on properties of a specific theory (classical, quantum,...) but on device replication and independence



Proof by contradiction

$$P_{\text{inf}}^{\text{AC}} = P_{\text{GHZ}}^{\text{AC}} = \frac{1}{2} ([00] + [11])$$

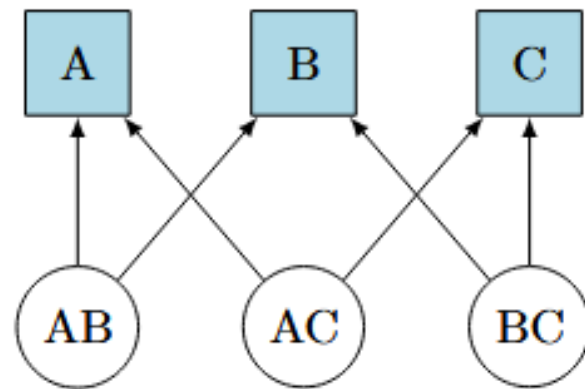
$$P_{\text{inf}}^{\text{BC}} = P_{\text{GHZ}}^{\text{BC}} = \frac{1}{2} ([00] + [11])$$

$$P_{\text{inf}}^{\text{AB}} = P_{\text{inf}}^{\text{A}} P_{\text{inf}}^{\text{B}}$$

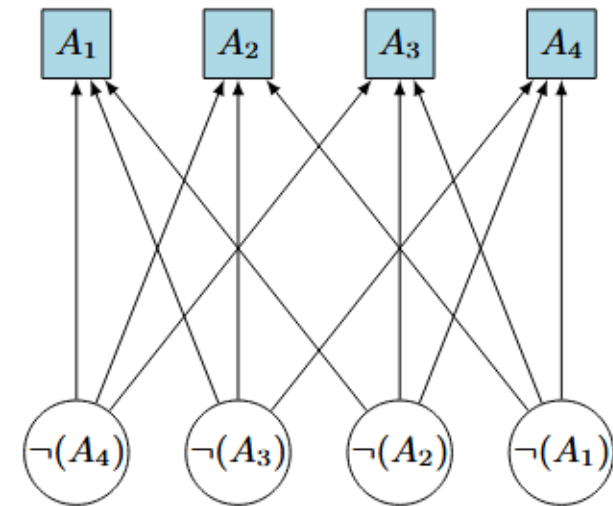


Shared Randomness vs. Common Cause

Is a shared random bit possible without a common cause?



b
Triangle scenario

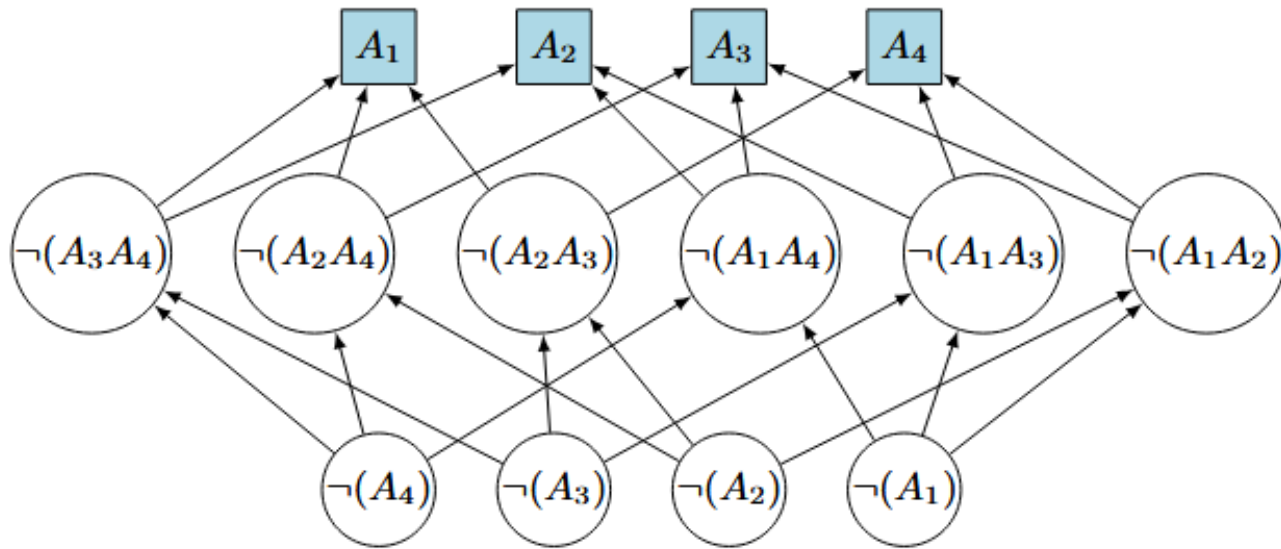


c
One-layer tetrahedron scenario

In networks without intermediate layers, the answer is no!

Shared Randomness vs. Common Cause

Is a shared random bit possible without a common cause?



d
Full two-layer tetrahedron scenario

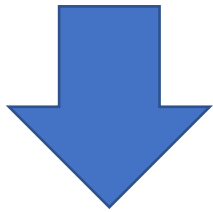
More complicated structure, cannot be reduced to a single-layer scenario in general



No cloning of information

Shared Randomness vs. Common Cause

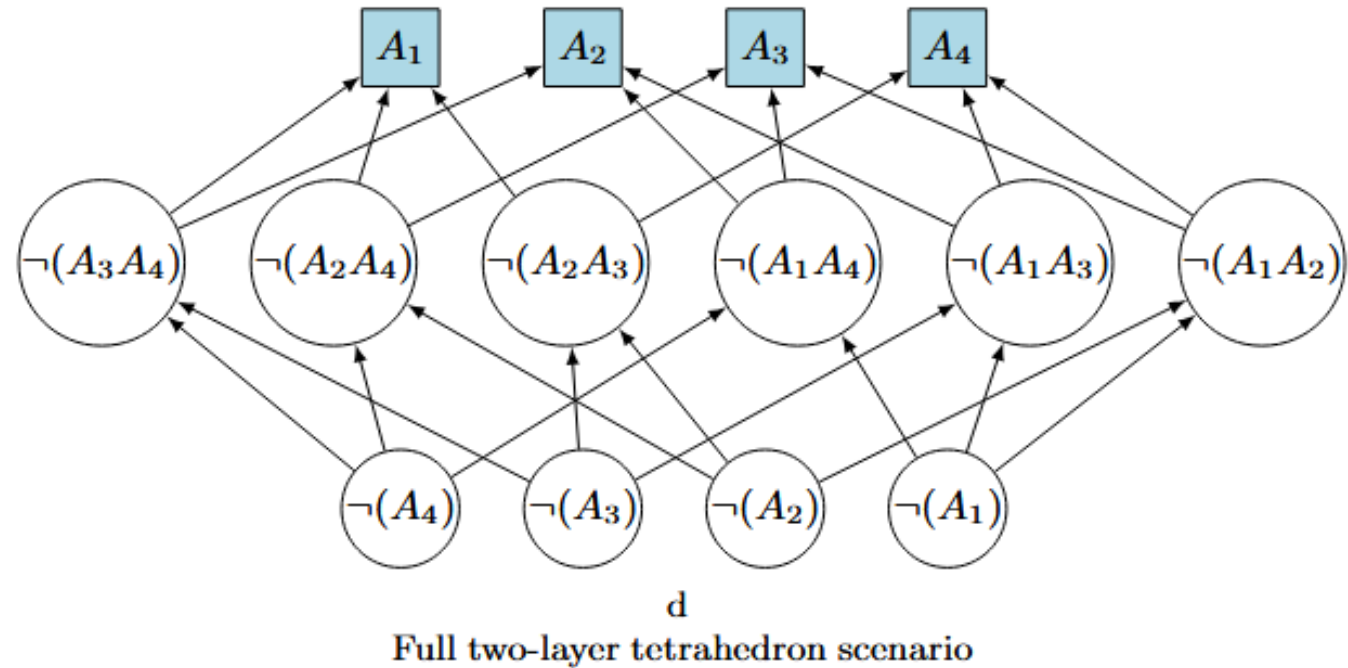
Just use inflation again?!



No non-trivial inflation without copying information here!



Classical infeasibility but what about quantum and no-signaling?



Shared randomness beyond Quantum

No non-trivial inflation, so what can we say?



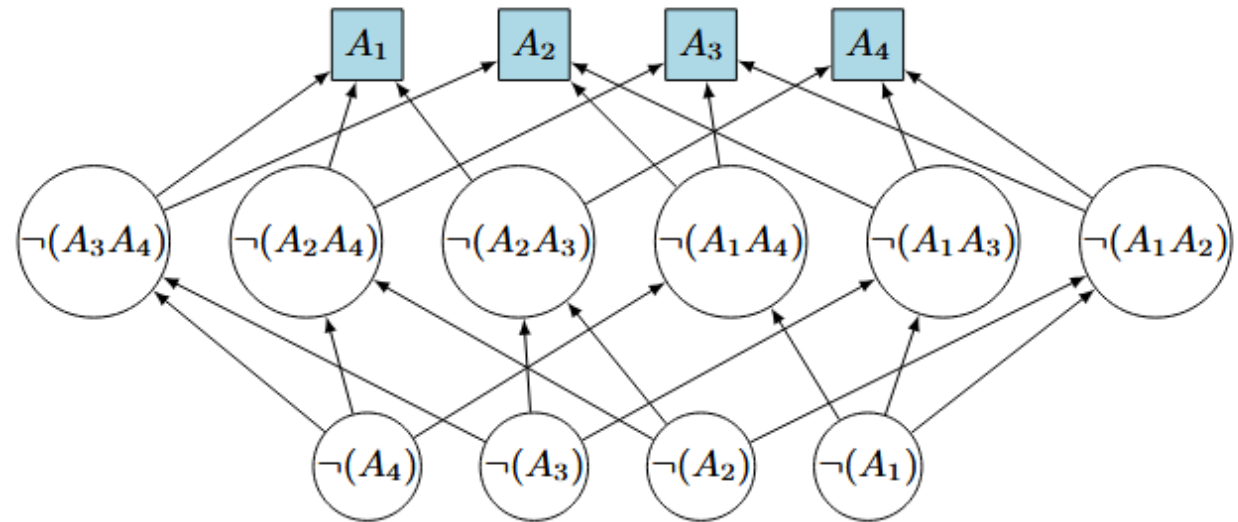
No constraints, so SR-bit should be possible?



We find a toy theory that implements the SR-bit



It is causal, no-signaling, allows replication of devices



d

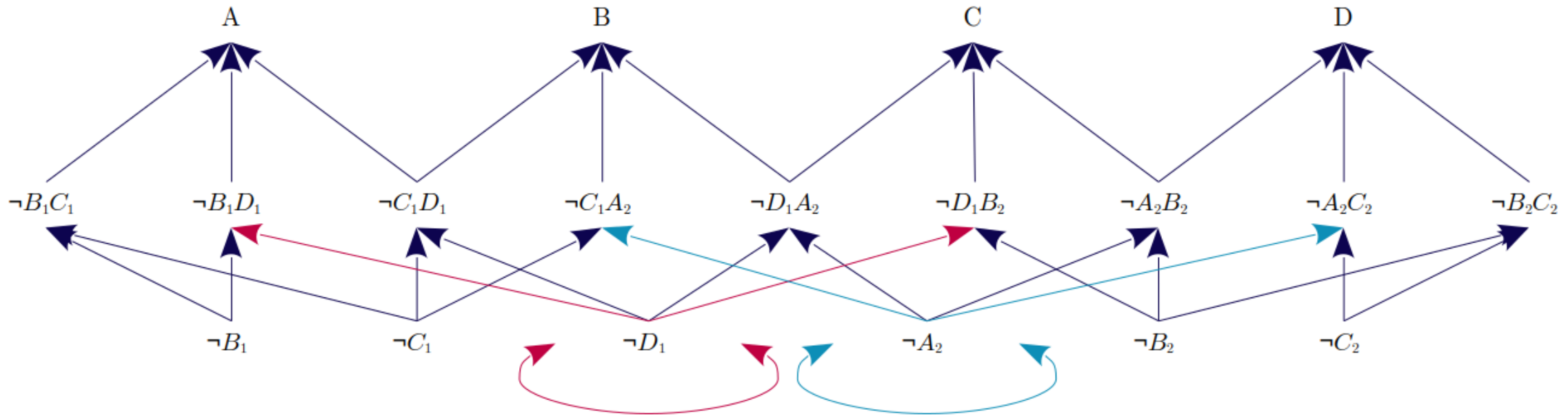
Full two-layer tetrahedron scenario

Shared randomness in Quantum Theory

No non-trivial inflation, so what can we say?

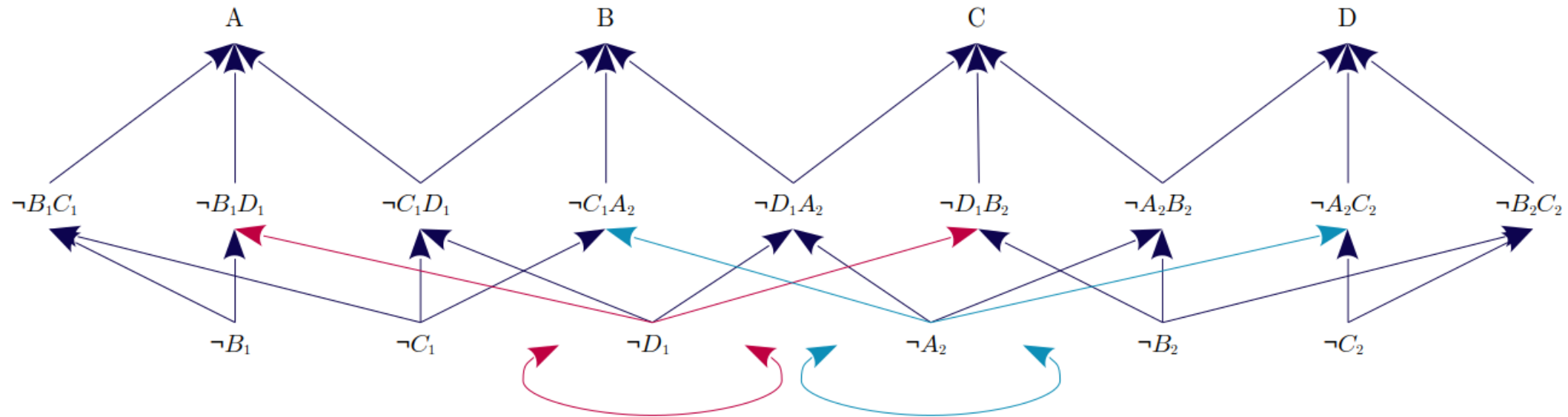


We have the quantum formalism available



Unphysical scenario but it leads to mathematical constraints

Shared randomness in Quantum Theory



Define:

$$|\psi_{\mathcal{I}}\rangle := |\neg B_1\rangle |\neg C_1\rangle |\neg D_1\rangle |\neg A_2\rangle |\neg B_2\rangle |\neg C_2\rangle$$

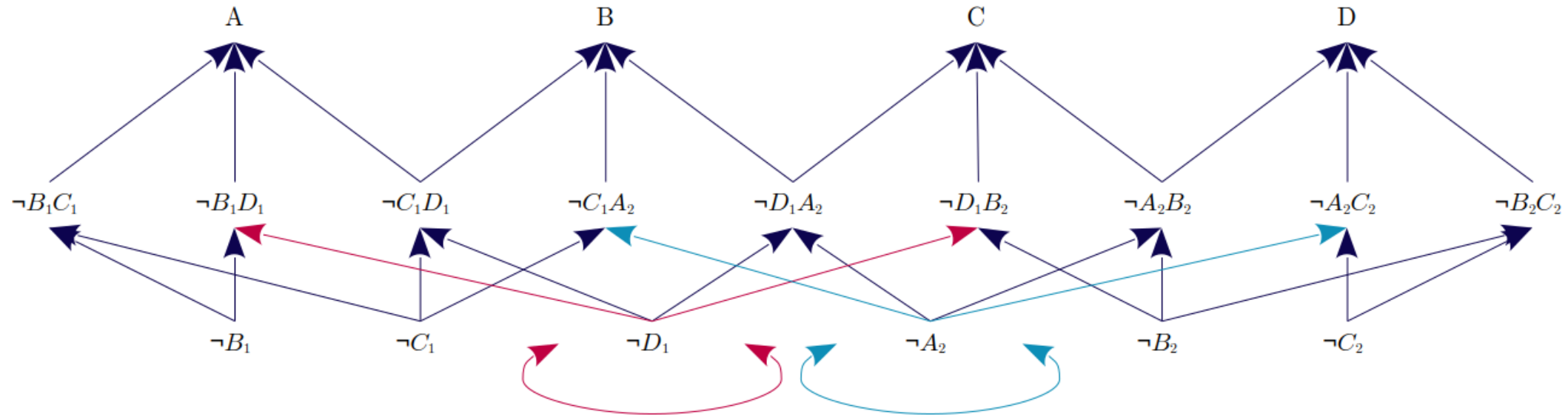
$$\tilde{A}_0 := U_{\neg B_1 C_1}^\dagger U_{\neg B_1 D_1}^\dagger U_{\neg C_1 D_1}^\dagger A_0 U_{\neg B_1 C_1} U_{\neg B_1 D_1} U_{\neg C_1 D_1}$$



$$\begin{aligned} [\tilde{A}_0, \tilde{B}_0] &= 0, \\ [\tilde{B}_0, \tilde{C}_0] &= 0, \\ [\tilde{C}_0, \tilde{D}_0] &= 0, \end{aligned}$$

But no commutation between A and C, and B and D

Shared randomness in Quantum Theory



From inflation:

$$\langle \psi_{\mathcal{I}} | \tilde{A}_0 \tilde{B}_0 | \psi_{\mathcal{I}} \rangle = p(A = 0, B = 0)$$

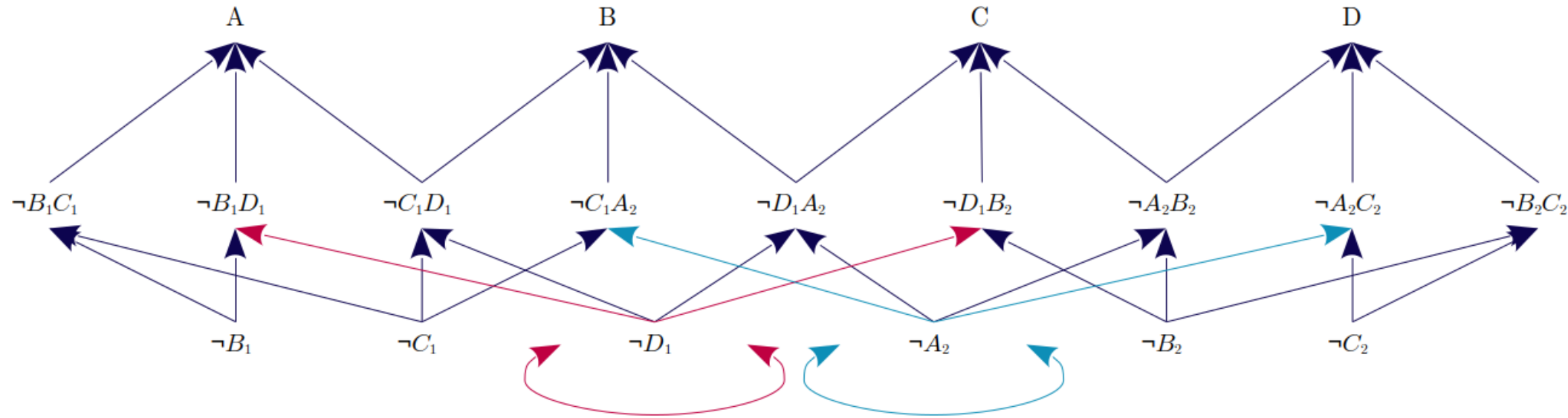
$$\langle \psi_{\mathcal{I}} | \tilde{A}_0 | \psi_{\mathcal{I}} \rangle = p(A = 0)$$

$$\langle \psi_{\mathcal{I}} | \tilde{B}_0 | \psi_{\mathcal{I}} \rangle = p(B = 0)$$

Assume:

$$p(A = 0, B = 0) = p(A = 0) = p(B = 0) = p(D = 0) = 1/2.$$

Contradiction via SOS



Hence

$$\tilde{A}_0 |\psi_I\rangle = \tilde{B}_0 |\psi_I\rangle$$

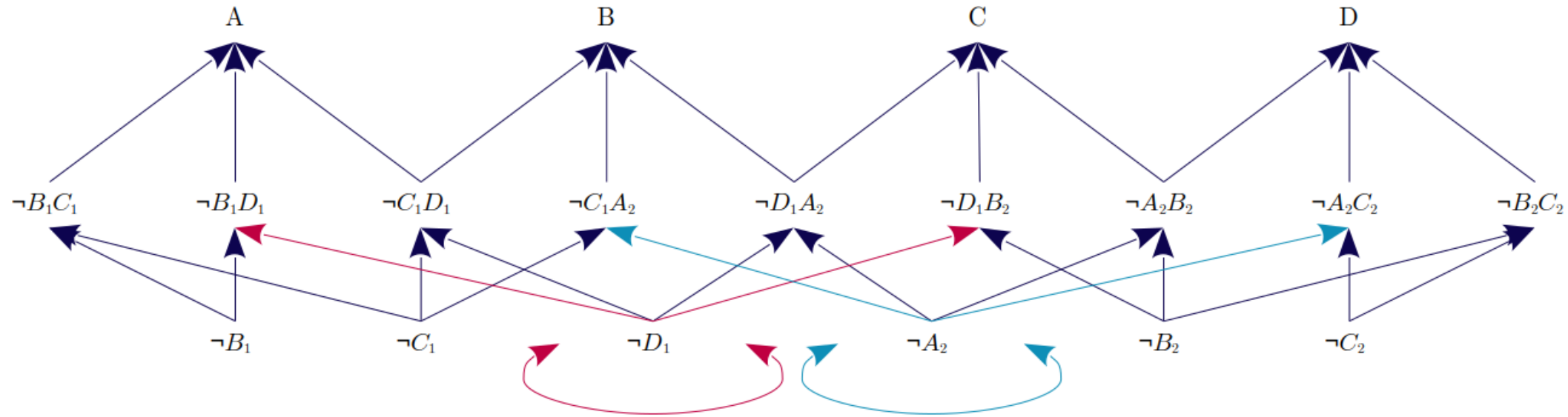
$$\tilde{B}_0 |\psi_I\rangle = \tilde{C}_0 |\psi_I\rangle$$

$$\tilde{C}_0 |\psi_I\rangle = \tilde{D}_0 |\psi_I\rangle$$

$$\tilde{A}_0 |\psi_I\rangle = \tilde{D}_0 |\psi_I\rangle$$



Contradiction via SOS



$$\tilde{A}_0 |\psi_{\mathcal{I}}\rangle = \tilde{D}_0 |\psi_{\mathcal{I}}\rangle$$

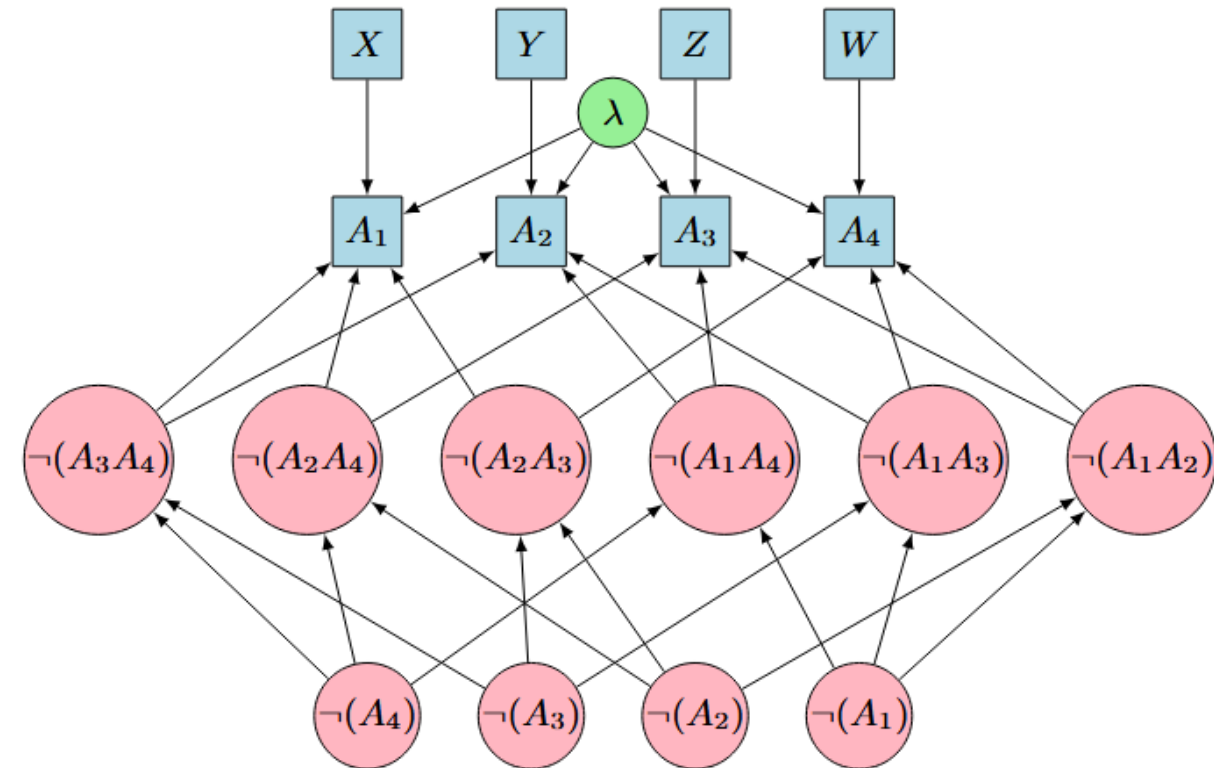
$$\begin{aligned} \langle \psi_{\mathcal{I}} | \tilde{A}_0 \tilde{D}_0 | \psi_{\mathcal{I}} \rangle &= \langle \psi_{\mathcal{I}} | \tilde{A}_0^2 | \psi_{\mathcal{I}} \rangle \\ &= \langle \psi_{\mathcal{I}} | \tilde{A}_0 | \psi_{\mathcal{I}} \rangle \\ &= p(A = 0) \\ &= 1/2 \end{aligned}$$

But A and D are independent

$$\begin{aligned} \langle \psi_{\mathcal{I}} | \tilde{A}_0 \tilde{D}_0 | \psi_{\mathcal{I}} \rangle &= \langle \psi_{\mathcal{I}} | \tilde{A}_0 | \psi_{\mathcal{I}} \rangle \langle \psi_{\mathcal{I}} | \tilde{D}_0 | \psi_{\mathcal{I}} \rangle \\ &= p(A = 0) p(D = 0) \\ &= 1/4 \end{aligned}$$

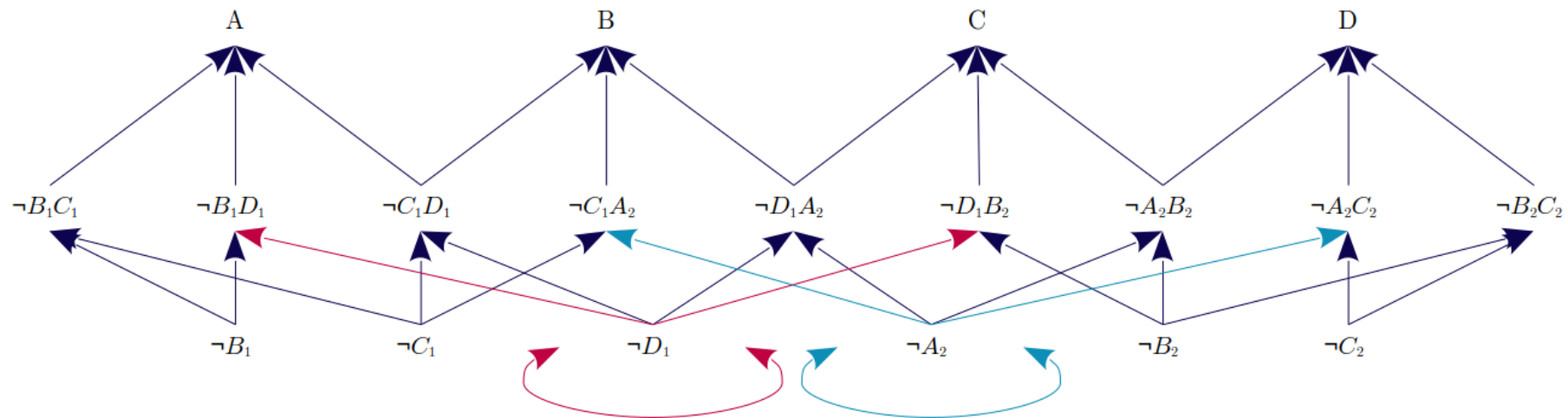
Extensions

- We show this Quantum vs. Post-Quantum separation for every $n \geq 4$ (with more layers)
- We provide a quantitative version of our proof (inequality/game)
- We extend the reasoning to the GHZ state



Summary

- Separation between Quantum and post-quantum theory in a multi-layer circuits
- Analytical proof that is scalable for every $n \geq 4$
- Uses SOS and inflation arguments
- Leads to certificates in terms of games/inequalities



The End